

UNIVERSAL LLC CONTROLLER

Firmware Reference Manual

Revision	Date	Firmware Version	Author	Description
01	07-10-2026	v1.0	Behjati H.	First draft



Contents

1 Introduction 3

2 Steady State Operation 4

2.1 First Harmonic Approximation (FHA) 4

2.2 Steady State Voltage Curves..... 6

3 Dynamic Model 10

3.1 Frequency Modulation 10

3.2 Duty Cycle Modulation..... 12

4 Controller Design 14

4.1 Nonlinear Double Actuator Control 14

4.2 Closed Loop Response 15

5 Synchronous Rectification 17

6 State Machine 18

7 Parameters Dictionary 19

1 Introduction

This document constitutes the Firmware Reference Manual for Infineon's "Universal LLC Controller" firmware. The presented LLC controller is universal because it supports various permutations of LLC Type¹ and Control Mode², as seen in the Table 1 below.

Table 1: Various permutations of LLC Type and Control Mode.

LLC Type ¹	Control Mode ²
Half Bridge	Voltage Control
Half Bridge	Frequency Control
Half Bridge	Duty Cycle Control
Full Bridge	Voltage Control
Full Bridge	Frequency Control
Full Bridge	Duty Cycle Control

- **LLC Type¹** is selected via the parameter `params.llc.nom.type`.
- **Control Mode²** is selected via the parameter `params.ctrl.id.mode`.

The presented LLC controller is hardware agnostic and can work across multiple hardware platforms with different microcontroller families and power boards.

The remaining chapters of this document present steady state operation, dynamic model, controller design, synchronous rectification, state machine, and parameters dictionary.

The last chapter (parameters dictionary) provides a one-to-one mapping between the materials provided in this document and their corresponding parameters in the firmware.

2 Steady State Operation

The steady state operation of resonant converters, including LLC topology, can be analyzed by using the First Harmonic Approximation (FHA) method.

2.1 First Harmonic Approximation (FHA)

Figure 1 depicts a typical half-bridge LLC topology. Important voltages and currents can be seen on the circuit schematic. The voltages v_{sw} , v_{oe} , v_{o1} , and v_{o2} are of square shape with a frequency equal to the switching frequency. The output voltage, v_o , is DC (assuming adequate output capacitance). The currents i_{oe} and i_m are of sine shape and triangular shape, respectively, while i_r is the sum of i_{oe} and i_m .

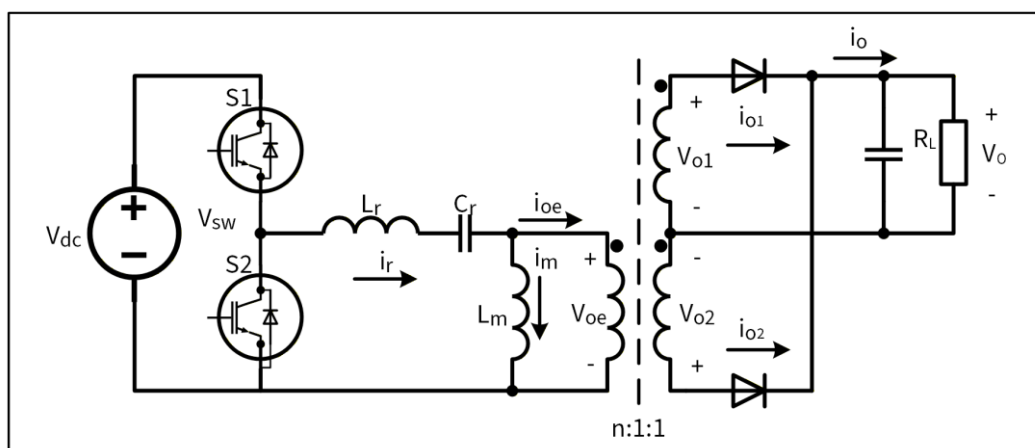


Figure 1: Typical half-bridge LLC topology

It is obvious that the combination of L_r and C_r would be a short circuit at series resonant frequency, $f_s = 1/(2\pi\sqrt{L_r \cdot C_r})$ i.e. the fundamental component of v_{sw} will be directly applied to the transformer at f_s . On the other hand, at very low frequencies, C_r , will block the current path and at very high frequencies L_r will block the current path.

The main premise of FHA method is that since the converter passes the input signal through efficiently around the resonant frequency f_s and blocks the signals that are not close to f_s , only fundamental AC components of the voltages and currents can be considered for steady state analysis i.e. all square waveforms can be replaced by their fundamental AC component and all pulsating DC signals can be replaced by their equivalent DC component. Figure 2 below illustrates this concept.

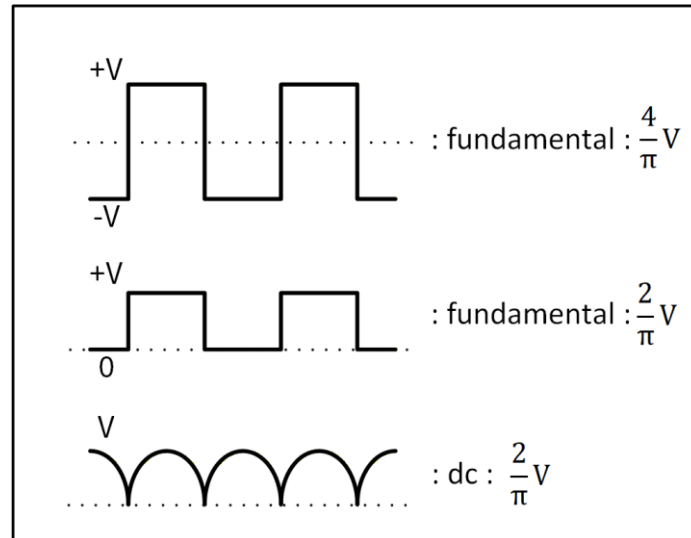


Figure 2: FHA and DC equivalents of various waveforms seen in LLC converters

It should be noted that a full bridge topology is like a half-bridge topology except that the fundamental component of the switching voltage will be twice that of the half-bridge topology, as shown in Figure 2.

Using the FHA method, the fundamental AC components of v_{oe} and i_{oe} can be expressed in terms of DC values of v_o and i_o . Consequently, an equivalent resistance R_e can be found as follows:

$$R_e = \frac{v_{oe}}{i_{oe}} = \frac{n \cdot \frac{4}{\pi} \cdot v_o}{\frac{1}{n} \cdot \frac{\pi}{2} \cdot i_o} = \frac{8}{\pi^2} n^2 R_L \quad (1)$$

Considering this equivalent resistance, an equivalent FHA model of the half-bridge LLC circuit can be derived where R_e represents an artificial resistance whose power consumption is transferred to the output port (as modeled by a power source illustrated below).

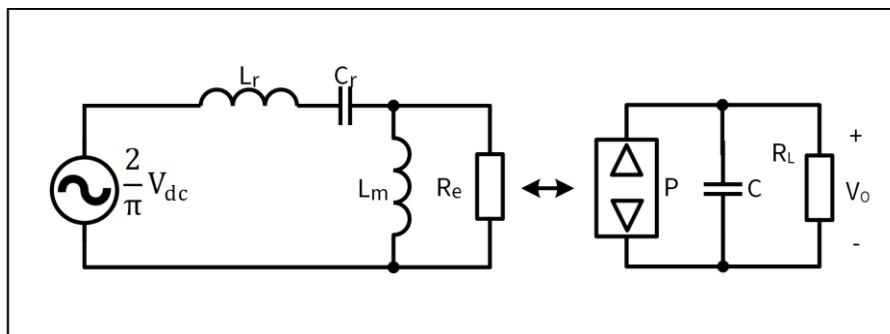


Figure 3: Equivalent FHA model of a typical half-bridge LLC converter

The equivalent FHA model in Figure 3 can be analyzed to obtain the input to output voltage gain of the LLC converter in steady state. To simplify the derivations, it is customary to perform some change of variables to “normalize” the switching frequency, f_{sw} , inductance, L_n , etc. as described below

$$\left\{ \begin{array}{l} f_n = \frac{f_{sw}}{f_s} \\ L_n = \frac{L_m}{L_r} \\ f_s = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{L_r \cdot C_r}} \\ f_p = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{(L_r + L_m) \cdot C_r}} \\ Q_e = \frac{\sqrt{L_r/C_r}}{R_e} \end{array} \right. \quad (2)$$

Given the change of variables defined in (2), the voltage gain of the LLC converter can be expressed in terms of normalized frequency, f_n , quality factor, Q_e , and normalized inductance, L_n :

$$M(f_n, Q_e, L_n) = \frac{1 \text{ or } 1/2 \text{ (Full vs. Half Bridge)}}{n \cdot \sqrt{\left(1 + (1/L_n)(1 - 1/f_n^2)\right)^2 + Q_e^2(f_n - 1/f_n)^2}} \quad (3)$$

Next section depicts the voltage gain curves of a typical LLC converter.

2.2 Steady State Voltage Curves

As shown in (3), the voltage gain of an LLC converter is a function of normalized frequency, f_n , quality factor, Q_e , and normalized inductance, L_n . Here, a typical full-bridge LLC converter with the following parameters is considered to plot the voltage curves versus switching frequency for various output loads:

$$\{v_{dc} = 40V, v_{out} = 5V, n = 7, L_r = 2.6\mu H, L_m = 12.5\mu H, C_r = 36nF, C = 1690\mu F\}.$$

The resulting output voltage curves for a wide range of output loads $P_{out} = 50W, 75W, \dots, 200W$ are depicted in Figure 4.

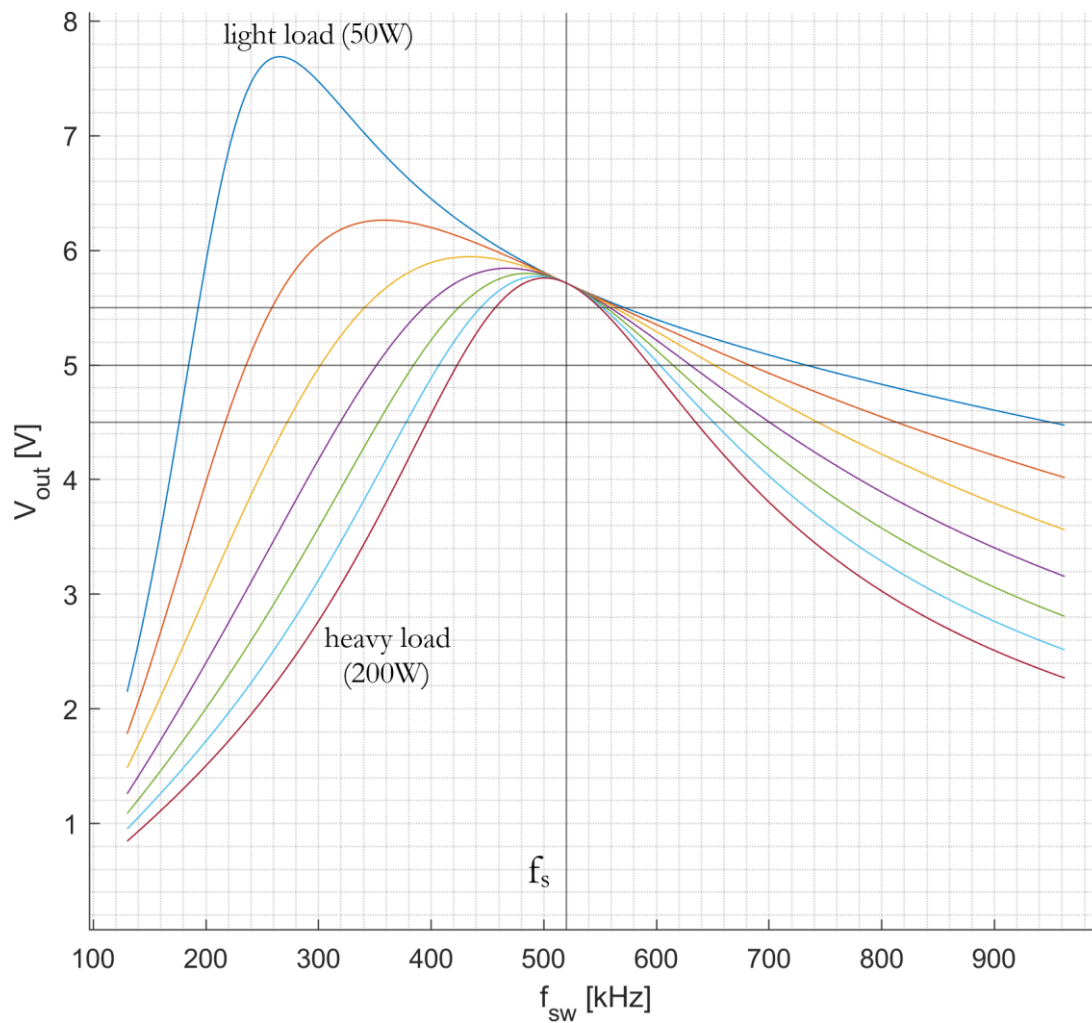


Figure 4: Typical output voltage curves of a typical LLC converter

As can be seen in Figure 4, in order to regulate the output voltage at 5V, a regulation band of 4.5V-5.5V can be considered and the corresponding switching frequency range based on the output load can be extracted from the output voltage curves. The lower the output load, the higher the switching frequency that is required to bring down the output voltage. Sometimes in LLC converters, the output voltage cannot be brought down within an acceptable switching frequency range, and a concurrent duty cycle modulation may be required to regulate the output voltage at very light output loads or when the output voltage command is lower than what the LLC transformer can provide. This topic will be discussed further in section 3.2 (Duty Cycle Modulation).

Usually, LLC converters are designed to operate above the series resonant frequency, f_s , because above f_s the converter always operates at Zero Voltage Switching (ZVS), no matter the output load. This is because above f_s , the resonant current i_r always lags behind the fundamental component of the switching voltage v_{sw} . However, below f_s , the converter may or may not operate at ZVS depending on the output load.

Figure 5 shows a zoomed-in view of Figure 4 above the series resonant frequency where the operating range of the switching frequency should be. As can be seen in Figure 5, an operating switching frequency range of ~540kHz to ~960kHz is required to cover the voltage regulation from light to heavy output load conditions.

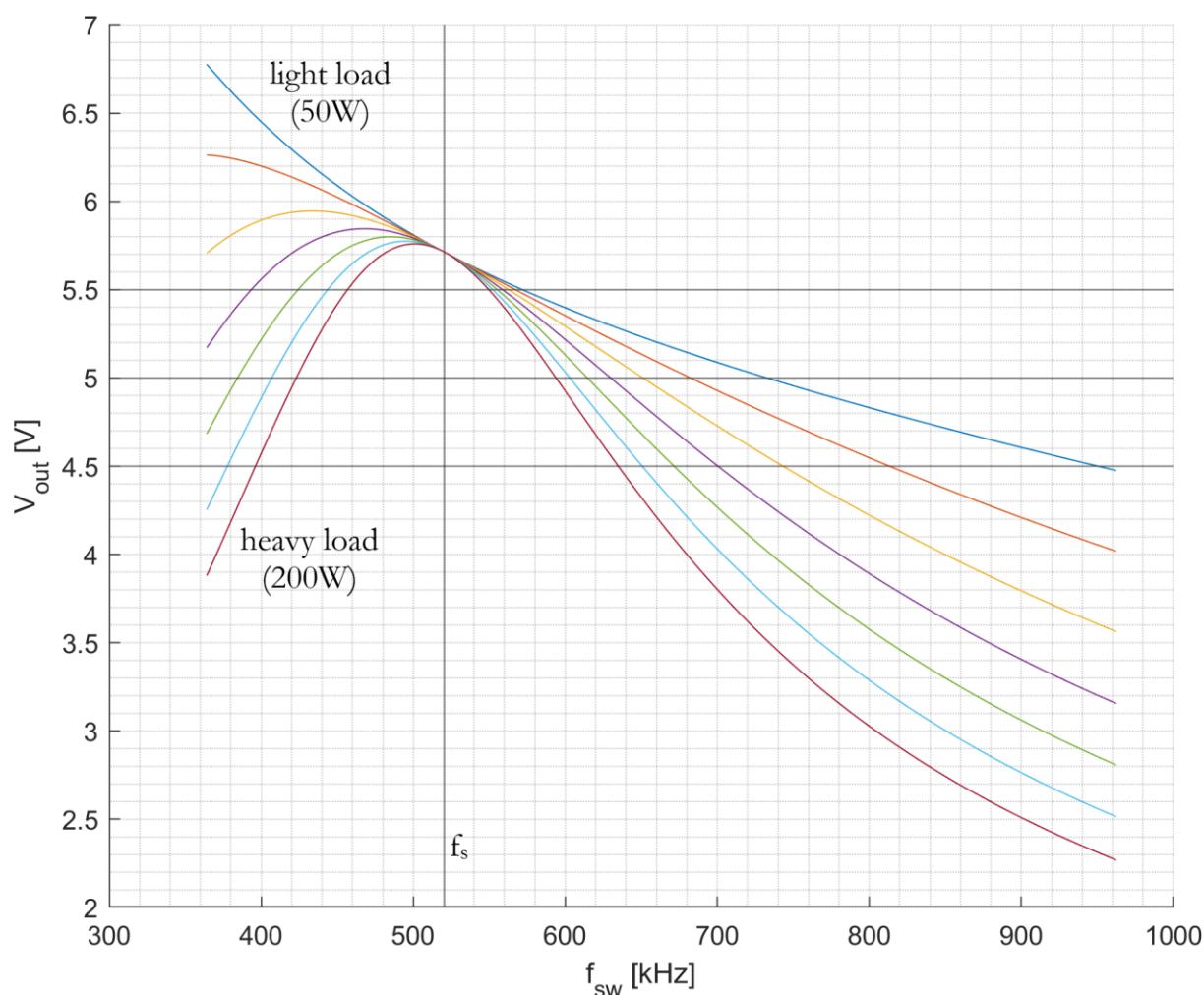


Figure 5: Frequency operation range of a typical LLC converter above the series resonant frequency

Another thing to notice from Figure 5 is that increasing the switching frequency decreases the output voltage which indicates a negative slope. Usually the switching period, T_{sw} , is controlled by the microcontroller instead of the switching frequency, f_{sw} , because the PWM generators accept the period not the frequency. This also has the added benefit of creating output voltage curves with positive slopes as illustrated in Figure 6, hence, the feedback error can be fed to the PI controller directly without having to do polarity inversion (to avoid instability due to positive feedback).

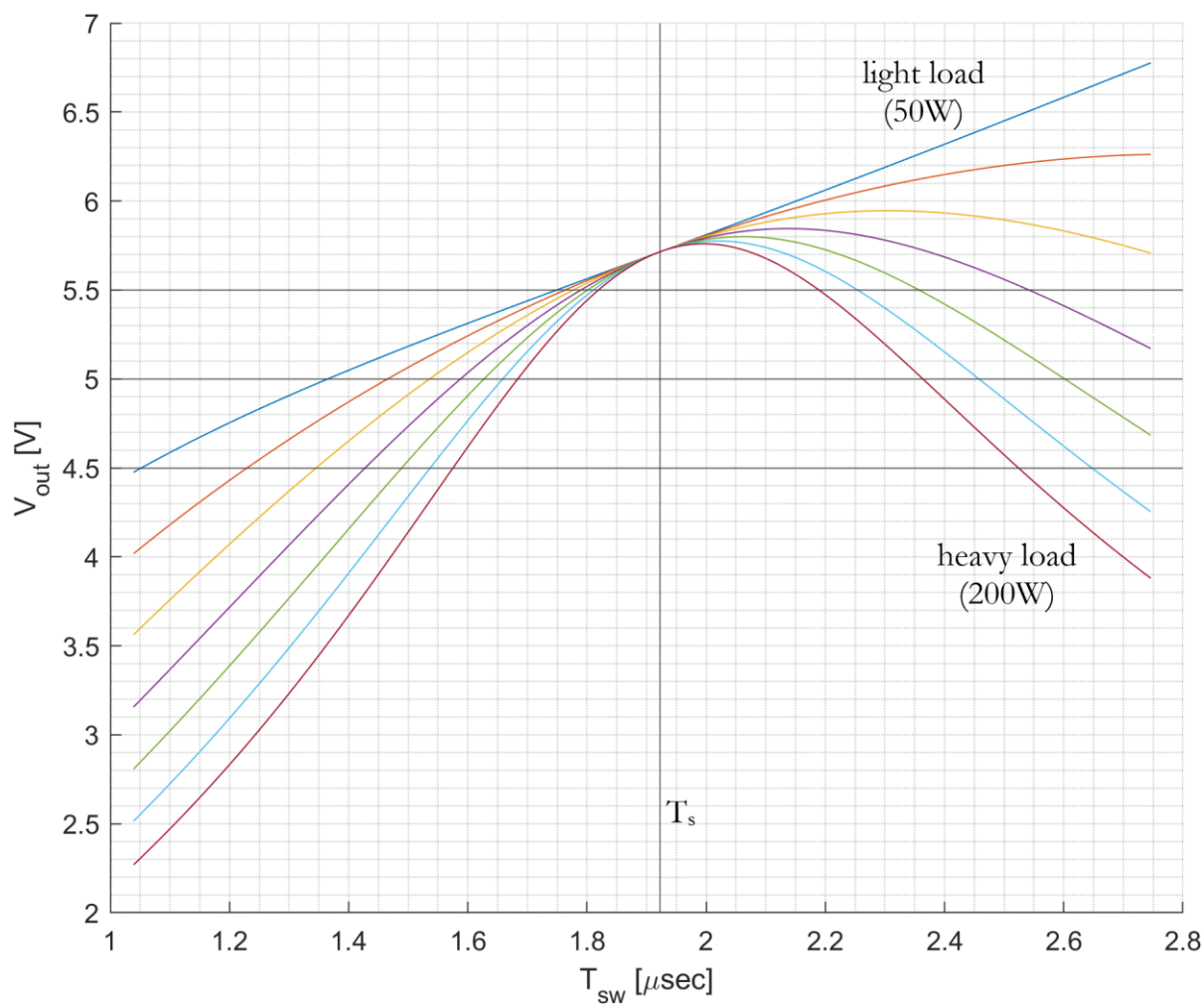


Figure 6: Switching period operation range of a typical LLC converter below the series resonant switching period

3 Dynamic Model

The objective of this section is to develop dynamic models to represent the plant when modulating either switching frequency or duty cycle. As mentioned in previous section, usually LLC converters are controlled by frequency modulation, especially at full load. However, in light load conditions or when the voltage command is much lower than the nominal output voltage, duty cycle modulation may be necessary to be able to bring down the voltage. The dynamic models developed in this section will be the basis for designing the controller in section 4.

3.1 Frequency Modulation

As described in section 2.1, v_{oe} and i_{oe} can be replaced by their equivalent fundamental AC components to obtain an equivalent circuit with dependent voltage and current sources for v_{oe} and i_{oe} as seen in Figure 7 below.

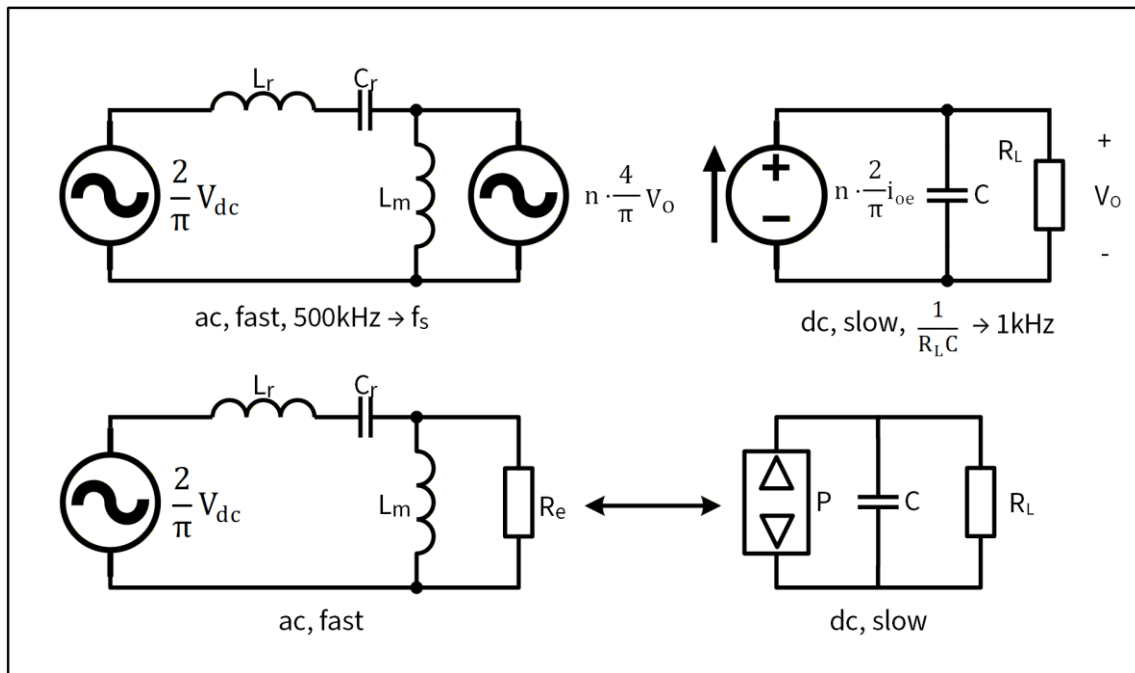


Figure 7: Separation of fast and slow dynamics in a typical half-bridge LLC converter

Here, we can write R_e in terms of v_o similar to (1) again, except that the DC value for i_o must also include the dynamic current going through the output capacitor i.e. $C dv_o/dt$. It must be noted that this current is zero in steady state which is why it was discarded in (1). Therefore, dynamic R_e would be

$$R_e = \frac{v_{oe}}{i_{oe}} = \frac{n \cdot \frac{4}{\pi} \cdot v_o}{\frac{1}{n} \cdot \frac{\pi}{2} \cdot (v_o/R_L + C dv_o/dt)} = \frac{8}{\pi^2} n^2 \cdot \frac{R_L}{1 + \frac{R_L C}{v_o} \cdot \frac{dv_o}{dt}} \quad (4)$$

There is a fundamental assumption here that the output capacitor is very large, i.e. the secondary-side circuit (right side of Figure 7) has a very large time constant compared to the primary-side circuit (left side of Figure 7). This is also required so that the output voltage ripple across the load, which is determined by the capacitor size, is minimized. Therefore, from the perspective of the primary side (left circuit in Figure 7), we can approximate $R_e = 8n^2 R_L / \pi^2$. However, on the secondary side, the dynamics of the capacitor must be accounted for by considering the capacitor current $C dv_o/dt$.

The output power balance equation can be written as

$$p = v_o i_o = v_o \left(C \frac{dv_o}{dt} + \frac{v_o}{R_L} \right) \quad (5)$$

Applying small-signal perturbation and linearization around the operating point, i.e. $x = X + \hat{x}$, equation (5) will result in

$$\hat{p} = CV_o \frac{d\hat{v}_o}{dt} + 2 \frac{V_o}{R_L} \hat{v}_o \quad (6)$$

which can be transformed into Laplace domain as

$$\hat{p} = V_o \left(Cs + \frac{2}{R_L} \right) \hat{v}_o = V_o \cdot \frac{(R_L C/2)s + 1}{R_L/2} \hat{v}_o \quad (7)$$

Therefore,

$$\hat{v}_o = \frac{1}{V_o} \cdot \frac{R_L/2}{(R_L C/2)s + 1} \hat{p} \quad (8)$$

Equation (8) describes how perturbations in the power source in Figure 7 affect the output voltage. However, we want to find out the effect of perturbation in the switching frequency on the output voltage. Thus, let us express $\hat{p}$ in terms of $\hat{f}_s$ as follows:

$$\hat{p} = \frac{\partial p}{\partial f_s} \cdot \hat{f}_s = \frac{\partial (v_o^2/R_L)}{\partial f_s} \cdot \hat{f}_s = \frac{2V_o}{R_L} \cdot \frac{\partial v_o}{\partial f_s} \cdot \hat{f}_s \quad (9)$$

Combining (7) and (9), one can obtain

$$V_o \cdot \frac{(R_L C/2)s + 1}{R_L/2} \hat{v}_o = \frac{2V_o}{R_L} \cdot \frac{\partial v_o}{\partial f_s} \cdot \hat{f}_s \quad (10)$$

In other words,

$$\hat{v}_o = \frac{\frac{\partial v_o}{\partial f_s}}{\frac{R_L C}{2}s + 1} \cdot \hat{f}_s = \frac{\frac{\partial v_o}{\partial T_s}}{\frac{R_L C}{2}s + 1} \cdot \hat{T}_s \quad (11)$$

Equation (11) describes the dynamics of the plant in frequency modulation mode i.e. the effect of switching frequency (or switching period) perturbations on the output voltage.

It must be noted that it is more convenient to consider switching period for control purposes as explained in section 2.2 because the PWM generators inside the microcontroller accept period rather than frequency as the input and the slope of the voltage curve is positive when modulating the switching period:

$$\begin{cases} K_f = \left. \frac{\partial v_o}{\partial f_s} \right|_{v_o=V_o} < 0 \\ K_T = \left. \frac{\partial v_o}{\partial T_s} \right|_{v_o=V_o} > 0 \end{cases} \quad (12)$$

Figure 8 depict the typical voltage gain curves and slopes when modulating the switching frequency vs. the switching period.

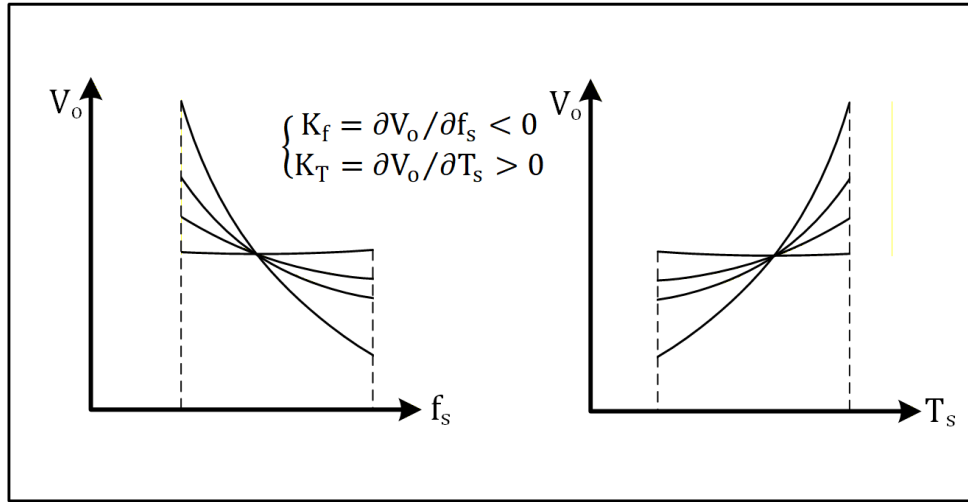


Figure 8: Voltage gain slopes corresponding to controlling frequency or switching period

3.2 Duty Cycle Modulation

Duty-cycle modulation in the context of LLC converters means modulating the duration of on/off power pulses. This is also known as burst mode. In other words, the duty cycle of the square wave switching voltage is always 50% to be able to create a fundamental AC component. However, there is also a much lower frequency i.e. the burst frequency which determines the frequency of on/off pulsating behavior of the converter in duty-cycle control mode. As an example, consider that the switching frequency is 500kHz, while the burst frequency is 10kHz and duty cycle is 20%. This means the PWM generator generates 10 periods of 500kHz switching signal followed by turning off for 40 periods of 500kHz, therefore creating a burst frequency of $500\text{kHz}/(40+10) = 10\text{kHz}$ and a burst duty cycle of $10/40=20\%$.

When modulating the duty-cycle, the switching frequency has already reached its maximum possible level for the LLC converter i.e. if there was no duty cycle modulation ($d = 1$), the converter would operate at $V_{o,min}$ corresponding to P_{min} , T_{min} , and f_{max} . Considering this, we can write the power balance in duty cycle control mode as

$$p = d \cdot P_{min} \rightarrow \hat{p} = \hat{d} \cdot P_{min} \quad (13)$$

Combining (8) and (13), one can express the effect of perturbations in duty cycle $\hat{d}$ on the output voltage $\hat{v}_o$ as follows:

$$\hat{v}_o = \frac{1}{V_o} \cdot \frac{\frac{R_L}{2}}{\left(\frac{R_L C}{2}\right)s + 1} (\hat{d} \cdot P_{min}) \quad (14)$$

$$= \frac{\left(\frac{R_L P_{min}}{2V_o}\right)}{\left(\frac{R_L C}{2}\right)s + 1} \hat{d}$$

Simplifying (14) would result in

$$\hat{v}_o = \frac{\left(\frac{V_{o,min}^2}{2V_o}\right)}{\left(\frac{R_L C}{2}\right)s + 1} \hat{d} \quad (15)$$

Equation (15) describes the plant dynamics model of the LLC converter in duty-cycle modulation mode. Similar to how K_f and K_T were defined in (11) and (12) for frequency modulation, we can define K_d here for duty-cycle modulation:

$$K_d = \frac{V_{o,min}^2}{2V_o} > 0 \quad (16)$$

By inspecting (11) and (15), it is evident that the plant dynamics model in both frequency and duty cycle control are similar except for the gains i.e. $\{K_f / K_T\}$ or K_d .

4 Controller Design

The purpose of this section is to design a robust controller that can always regulate the output voltage, no matter the output load or the voltage command. The proposed controller will switch seamlessly between frequency and duty cycle control. If the output voltage command can be achieved within the specified frequency range for the converter (the range is a configurable user-defined parameter), the controller converges to the required switching frequency within that range to regulate the voltage while keeping the duty cycle at 100%. However, if due to very light load or a low voltage command, the output voltage cannot be regulated by frequency control inside the user specified frequency range, it will automatically and seamlessly reduce the burst duty cycle at maximum switching frequency to regulate the output voltage.

Since the proposed controller adjusts both switching period and duty cycle as its outputs in a nonlinear fashion, it constitutes a “nonlinear double-actuator” controller. This concept will be explained in the next section.

4.1 Nonlinear Double Actuator Control

Figure 9 depicts the proposed nonlinear double actuator controller. The nonlinear functions $g_T(\tilde{v}_o)$ and $g_d(\tilde{v}_o)$ are defined in Figure 10.

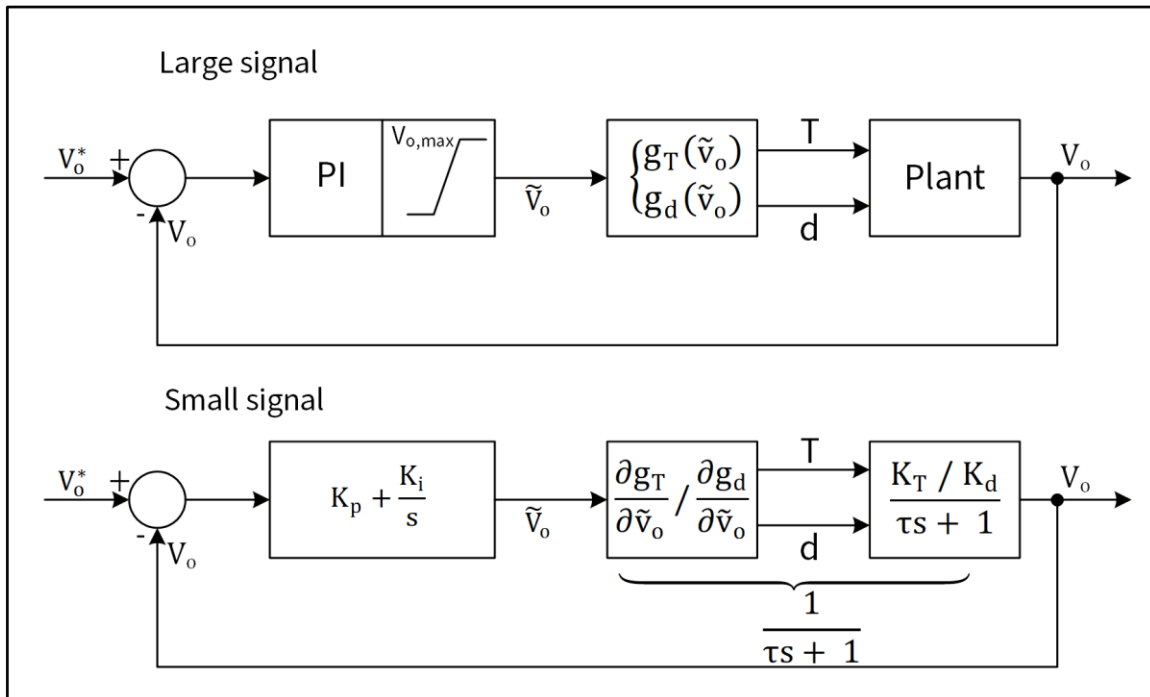


Figure 9: The nonlinear double-actuator controller: large-signal and small-signal models

As depicted in Figure 10, when the output of the PI controller, $\tilde{v}_o$ is less than $V_{o,min}$, the switching period stays at minimum (i.e. maximum switching frequency), while the duty cycle modulation starts in a nonlinear and parabolic fashion.

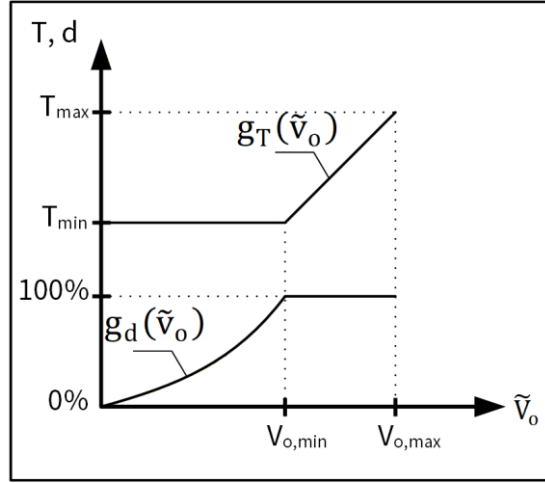


Figure 10: The nonlinear double-actuator controller: switching-period and duty-cycle functions

The analytical equations of the switching-period and duty-cycle functions in Figure 10 can be written as

$$g_T(\tilde{v}_o) = \begin{cases} T_{min} & : \tilde{v}_o < V_{o,min} \\ T_{max} & : \tilde{v}_o \geq V_{o,max} \\ \left(\frac{T_{max} - T_{min}}{V_{o,max} - V_{o,min}} \right) (\tilde{v}_o - V_{o,min}) + T_{min} & : \text{otherwise} \end{cases} \quad (17)$$

and

$$g_d(\tilde{v}_o) = \begin{cases} 1 & : \tilde{v}_o \geq V_{o,min} \\ \left(\frac{\tilde{v}_o}{V_{o,min}} \right)^2 & : \tilde{v}_o < V_{o,min} \end{cases} \quad (18)$$

The design of the parabolic function in (18) is not random. It is designed as a parabolic function to make the dynamics of the controller / plant combination in both frequency and duty-cycle modes identical and hence make the transitions and dynamic responses seamless and smooth. Next section elaborates on why the parabolic function in (18) will unify the mentioned dynamics.

4.2 Closed Loop Response

To develop the small-signal model of the closed-loop system, the slopes of $g_T(\tilde{v}_o)$ and $g_d(\tilde{v}_o)$ curves must be calculated as seen in Figure 9. The first slope, $\partial g_T(\tilde{v}_o) / \partial \tilde{v}_o$ can be written as

$$\frac{\partial g_T(\tilde{v}_o)}{\partial \tilde{v}_o} = \left(\frac{T_{max} - T_{min}}{V_{o,max} - V_{o,min}} \right) \approx \frac{1}{K_T} \quad (19)$$

As proved by (19), the slope of $g_T(\tilde{v}_o)$ curve is always the inverse of the slope of the plant model in switching period control mode, K_T .

Similarly, the second slope $\partial g_T(\tilde{v}_o) / \partial \tilde{v}_o$ can be calculated by taking the derivative of (18) as follows:

$$\frac{\partial g_d(\tilde{v}_o)}{\partial \tilde{v}_o} = \left(\frac{2\tilde{v}_o}{V_{o,min}^2} \right) \quad (20)$$

At an arbitrary operating point $\tilde{v}_o = V_o$, the slope in (20) would be

$$\left. \frac{\partial g_d(\tilde{v}_o)}{\partial \tilde{v}_o} \right|_{\tilde{v}_o=V_o} = \left(\frac{2V_o}{V_{o,min}^2} \right) = \frac{1}{K_d} \quad (21)$$

Again, as proven by (21), the slope of $g_d(\tilde{v}_o)$ curve is always the inverse of the slope of the plant model in duty cycle control mode, K_d . This statement is independent of the operating point because as the operating point changes, the slope of $g_d(\tilde{v}_o)$ curve also changes because it is designed as a parabolic nonlinear function. Therefore, it will always compensate for K_d of the plant at any operating point.

To summarize,

$$\frac{\partial g_T(\tilde{v}_o)}{\partial \tilde{v}_o} \cdot K_T \approx 1 \quad (22)$$

and

$$\frac{\partial g_d(\tilde{v}_o)}{\partial \tilde{v}_o} \cdot K_d = 1 \quad (23)$$

Therefore, the loop gain derived from the small signal model in Figure 9 would always be

$$G_{ol}(s) = \left(K_p + \frac{K_i}{s} \right) \left(\frac{1}{\tau s + 1} \right) \quad (24)$$

no matter the control mode (switching period vs. duty cycle) or the operating point. This is the main advantage of the proposed nonlinear double-actuator controller proposed here.

The time constant τ in (24) is

$$\tau = \left(\frac{R_L \cdot C}{2} \right) \quad (25)$$

Therefore, designing K_p and K_i for the PI controller is very straight-forward for a given desired cross-over frequency by the user, ω_c :

$$\begin{cases} K_p = \omega_c \tau \\ K_i = \omega_c \end{cases} \quad (26)$$

Substituting (26) into (24) will result in a very well-behaved first-order loop-gain, no matter the control mode or the operating point, as:

$$G_{ol}(s) = \left(\frac{\omega_c}{s} \right) \quad (27)$$

Therefore, the close loop transfer function would be

$$G_{cl}(s) = \frac{G_{ol}(s)}{G_{ol}(s) + 1} = \left(\frac{1}{s/\omega_c + 1} \right) \quad (28)$$

which describes a smoothly damped first order loop with a bandwidth (or cross-over frequency) of ω_c .

5 Synchronous Rectification

Synchronous Rectification (SR) helps to reduce the conduction loss of the output rectifier by synchronous switching of the output switches. It is important that each switch is turned on only after the current is fully commutated to that switch, otherwise the operation of the LLC converter can be jeopardized.

There are multiple methods to ensure the SR switches turn on at the right time. Some methods rely on a direct signal generated by the secondary side of the hardware to trigger the SR switching. In lieu of such hardware signals, the estimated phase shift (i.e. time shift) between the switching voltage, v_{sw} , and transformer current, i_{oe} , must be employed to trigger the SR gating signals.

The transformer current, i_{oe} , can be expressed in steady state as

$$i_{oe} = \frac{v_{sw}}{R_e} \cdot \frac{L_n f_n^2}{(f_n^2 - 1)(1 + jL_n Q_e f_n) + L_n f_n^2} \quad (29)$$

Therefore, the phase lag between v_{sw} and i_{oe} would be

$$\theta = \angle \left(\frac{i_{oe}}{v_{sw}} \right) = -\tan^{-1} \left(\frac{(f_n^2 - 1)L_n Q_e f_n}{(f_n^2 - 1) + L_n f_n^2} \right) \quad (30)$$

It can be seen from (30) that for

$$f_n = 1 \Rightarrow \theta = 0 \quad (31)$$

i.e. at series resonant frequency, the phase lag is always zero, and the switching of the SR switches must coincide with switching of the primary side switches.

On the other hand, when $f_n \neq 1$ and at very heavy load,

$$R_L = 0 \Rightarrow Q_e = \infty \Rightarrow \theta = -\frac{\pi}{2} \quad (32)$$

i.e. the switching of the SR switches must be 90 degrees behind switching of the primary-side switches. In practice, the phase lag lies somewhere between 0 degrees and 90 degrees depending on the switching frequency and load.

The quality factor, Q_e , can be expressed in terms of the output load's conductance, G_L , as

$$Q_e = \frac{\sqrt{L_r/C_r}}{R_e} = \frac{\sqrt{L_r/C_r}}{\frac{8}{\pi^2} n^2} \cdot \frac{1}{R_L} = \frac{\sqrt{L_r/C_r}}{\frac{8}{\pi^2} n^2} \cdot G_L \quad (33)$$

Therefore, we can utilize a change of variables from $\{f_n, Q_e\}$ to $\{f_{sw}, G_L\}$ and save the required phase shift (or time shift) in a Look Up Table (LUT) in the firmware:

$$\theta(f_n, Q_e) = \theta(f_{sw}, G_L) \quad (34)$$

Firmware can estimate the output conductance since the output current and voltage are measured and monitored. The estimated output conductance can then be used as the second argument for the SR LUT.

6 State Machine

Figure 11 illustrates the state machine diagram of the Universal LLC Controller.

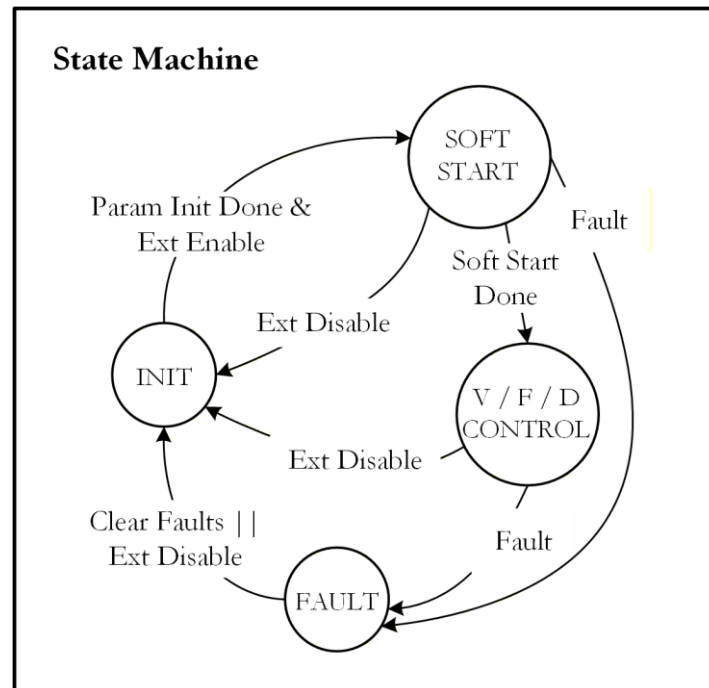


Figure 11: State machine diagram

The description of each state is listed in Table 2.

Table 2: States in the state machine.

State	Description
INIT	This is the first state by default where parameter initializations happen.
SOFT START	In this state, a high starting frequency (configurable as a parameter) is utilized for a specified amount of time (configurable as a parameter) to charge up the output capacitor in a controlled way and avoid current rush.
VOLTAGE CONTROL	In this state, the output voltage is regulated by the nonlinear double-actuator controller described in this document.
FREQUENCY CONTROL	In this state, the frequency is directly controlled as commanded by the user. This would allow the user to extract the voltage curves for tuning the frequency control mode. The duty cycle is always 1.0 here.
DUTY-CYCLE CONTROL	In this state, the duty cycle is directly controlled as commanded by the user. This would allow the user to extract the voltage curves for tuning the duty cycle control mode. The frequency is always the maximum specified frequency here.
FAULT	Fault state.

7 Parameters Dictionary

The following table presents a list of tunable firmware parameters. The firmware symbol, as well as the parameter type, its SI unit, the corresponding section in this document, and the corresponding mathematical symbol in that section are listed. This table provides a one-to-one mapping between the materials provided in this document and their corresponding parameters in the firmware.

There are two parameter types, Basic and Advanced. Advanced parameters don't need to be entered by the user. They can be auto-calculated by the firmware by invoking the auto-calculate function through the debugger or the GUI. The user can also set the advanced parameters directly if fine tuning is required. Usually, auto-calculation results in close-to-optimal results for initial evaluation but the advanced parameters can be further fine-tuned directly if desired.

In addition, not all basic parameters need to be touched for the control system to work. Only a few basic parameters usually need to be touched to get the system up and running. The firmware is very flexible and versatile, and all of the parameters can be finely tuned but that doesn't mean that they must be changed for the system to work.

Table 3: Parameters dictionary

FW Symbol	Type	Unit	Section	Symbol
TBD	B	-		
...				